

Disproof of Conjecture 000000007684

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Choose $q = 1/2$, which lies in the usual q -difference domain $0 < |q| < 1$. At the conjecture's listed scalar spectral value $\lambda = 0$, the eigenvalue equation is

$$D_q f(z) = 0, \quad \text{equivalently} \quad f(qz) = f(z).$$

Let the analytic function have power series $f(z) = \sum_{n \geq 0} c_n z^n$. Uniqueness of power-series coefficients gives

$$(q^n - 1)c_n = 0 \quad (n \geq 0).$$

For every $n \geq 1$, $q^n \neq 1$, so $c_n = 0$. Therefore f is constant. The boundary condition now says

$$0 = f(1) = c_0,$$

so f is the zero function. The scalar solution space at $\lambda = 0$ is therefore the zero vector space and has dimension 0, contradicting the asserted value $d(0) = 1$.

The Lean file works with coefficient sequences. For $q = 1/2$, the normalized positive-degree coefficient equation is $c_n = 0$ for $n > 0$; the boundary equation is $c_0 = 0$. Lean proves that every sequence satisfying both is identically zero, and that the zero sequence satisfies them. This exactly formalizes the algebraic coefficient step after the standard uniqueness theorem for analytic power series.